

A Tutor That Never Says “I Don’t Know”: Abstention Failure in a Retrieval-Augmented Question-Answering System and What It Means for Learners’ Trust

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Abstract

Retrieval-augmented generation (RAG) is often presented as a way to make language-model answers more reliable, which makes it an attractive design for AI tutors. Reliability for a learner, however, depends on more than accuracy. It also depends on whether the system signals when it has no good answer, because that signal is what allows a learner to calibrate trust. I built a RAG question-answering system from standard components (a DPR retriever, a FAISS index, and Flan-T5-base) and evaluated it on a seeded 500-question slice of SQuAD 2.0, of which 258 questions are labeled unanswerable. Retrieval improved exact match from 0.004 without retrieval to 0.062 with five retrieved passages. The system never abstained: it produced an answer string for all 258 unanswerable questions, and its answers looked the same whether or not retrieval had found the relevant passage, which happened for a minority of questions (Recall@10 = 0.310). I argue that this failure mode matters for education. A tutor that always answers removes the cue a learner would need to separate supported from unsupported output, and standard aggregate metrics do not reveal the problem. I discuss implications for how educational AI is evaluated and outline a study to test whether abstention helps learners rely on such systems appropriately.

Keywords: retrieval-augmented generation; abstention; unanswerable questions; trust calibration; appropriate reliance; AI in education

1. Introduction

Students increasingly ask AI systems questions they cannot yet answer themselves. That is exactly the situation in which they are least able to judge whether an answer is right. Lee and See (2004) argued that automation is often problematic because people fail to rely on it appropriately, and that trust guides reliance when a complete understanding of the automation is impractical. The goal is *appropriate* reliance, not maximal reliance: relying on a system when it is right and not when it is wrong. Relying on automation when one should not is misuse; failing to use it when it would help is disuse (Parasuraman & Riley, 1997). For a learner, appropriate reliance requires some signal of when the system is on solid ground.

Retrieval-augmented generation is a common response to the reliability problem. Instead of relying only on what a language model stores in its parameters, a RAG system retrieves passages from a document collection and conditions its answer on them (Lewis et al., 2020). This design is appealing for tutoring because the collection can be course material. But retrieval does not by itself tell the system, or the learner, when the retrieved material does not contain an answer.

The dataset used here makes that problem concrete. SQuAD 2.0 was built because reading-comprehension systems “tend to make unreliable guesses on questions for which the correct answer is not stated in the context,” and it requires systems to “determine when no answer is supported by the

paragraph and abstain from answering” (Rajpurkar et al., 2018, abstract; the paper calls the dataset SQuADRU, and it was released as SQuAD 2.0).

This paper reports what happened when I built a RAG system from standard, publicly available components and evaluated it on SQuAD 2.0. The system improved over a no-retrieval baseline, but it never once declined to answer. My question is what this failure mode means for learners:

If a tutor answers every question in the same way, whether or not it has a basis for the answer, what cue is left for a learner to calibrate trust?

The contribution is modest and specific. I document a clean instance of abstention failure in a small, reproducible RAG pipeline. I show why aggregate accuracy metrics hide it. I connect it to research on how people rely on language models, and I propose a study with learners. The system results are empirical. The education implications are arguments from prior literature, not findings of this study, and I label them as such.

2. Background

2.1 Abstention in question answering

A question-answering system can fail in two ways on a question it cannot answer: it can say so, or it can guess. SQuAD 2.0 added over 50,000 unanswerable questions, written adversarially by crowdworkers to look similar to answerable ones. A strong neural system that reached 86% F1 on the original SQuAD achieved only 66% F1 on SQuAD 2.0 (Rajpurkar et al., 2018).

Knowing when to abstain is harder when inputs differ from training data. Kamath et al. (2020) studied selective question answering under domain shift. They found that abstention policies based only on a model’s softmax probabilities fare poorly because models are overconfident on out-of-domain inputs. A separately trained calibrator did better: at 80% accuracy it answered 56% of questions, compared with 48% when using the model’s probabilities directly.

2.2 How people rely on language-model answers

Language models seldom signal doubt on their own. Zhou et al. (2024) found that deployed language models were reluctant to express uncertainty even when wrong, and were overconfident when prompted to state confidence. In the same work, users relied heavily on model outputs whether or not certainty was marked.

Users also overestimate model accuracy. Steyvers et al. (2025) found that people overestimated the accuracy of LLM answers when given default explanations. Longer explanations raised confidence without improving accuracy. They distinguish a *calibration gap* (the difference between people’s confidence in LLM answers and the model’s own confidence) from a *discrimination gap* (the difference between how well people and the model distinguish correct from incorrect answers). In fact-checking, LLM explanations made users more efficient than a search engine did, but users overrelied on the model when its explanation was wrong (Si et al., 2024).

Communicating uncertainty can help, but the form matters. In a pre-registered experiment, first-person expressions of uncertainty (“I’m not sure, but...”) reduced agreement with an LLM-infused search engine and increased accuracy, whereas uncertainty expressed from a general perspective (“It’s not clear, but...”) had weaker, non-significant effects (Kim et al., 2024). In an earlier decision-support study, showing confidence scores helped calibrate trust but did not by itself improve decisions (Zhang et al., 2020).

2.3 Learners as users

Learners are a special case because the goal is learning, not only a correct answer. In a randomized trial with high school mathematics students, unrestricted access to GPT-4 improved performance during practice but harmed performance once access was removed. A version with safeguards, including teacher-designed hints in place of answers, largely mitigated the harm (Bastani et al., 2025). In a lab study of 21 novice programmers, generative AI could compound the metacognitive difficulties of students who were struggling, and these students overestimated their performance (Prather et al., 2024). In a study of 432 undergraduates working on programming problems with an AI assistant, higher trust was associated with lower appropriate reliance in a non-linear relationship, and AI literacy moderated that relationship (Pitts et al., 2026).

Together these studies suggest that learners need cues about when AI output is supported. They also suggest that such cues cannot be assumed to exist by default.

3. System

I built the system as a course project (Section 10), following the retrieve-then-generate design of RAG (Lewis et al., 2020): encode the question, retrieve the top- K passages, and generate an answer conditioned on them. Unlike Lewis et al., who train the retriever and generator jointly and marginalize over passages, I concatenated the top- K passages into a single prompt. All components were used off the shelf, without fine-tuning.

Retriever. Dense Passage Retrieval (DPR) uses separate BERT-based encoders for questions and passages and scores passages by the dot product of their embeddings (Karpukhin et al., 2020). I used the publicly released facebook/dpr-question_encoder-single-nq-base and facebook/dpr-ctx_encoder-single-nq-base checkpoints, which were trained on Natural Questions.

Index. Passage embeddings were stored in a FAISS IndexFlatIP index, which performs exact maximum inner-product search (Johnson et al., 2021).

Generator. The generator was google/flan-t5-base, an instruction-finetuned T5 model released by Chung et al. (2024). The top- K retrieved passages were concatenated into one context and inserted into the prompt “*Answer the question. Context: {z} Question: {x} Answer:*”, truncated to 1,024 input tokens. Decoding used beam search with four beams and length normalization, with a maximum of 32 new tokens. The prompt contains no instruction about what to do when the context does not contain an answer.

Corpus. The retrieval collection consisted of approximately 20,000 unique passages that appear as contexts in SQuAD 2.0 (the validation split plus a seeded sample of the training split). They were split into windows of 120 tokens with a stride of 80, giving approximately 29,000 chunks. Because every evaluation question’s source paragraph is in the collection, a retrieval miss reflects the retriever, not missing coverage.

4. Evaluation

I evaluated on a fixed, seeded sample of 500 questions from the SQuAD 2.0 validation split: 242 answerable and 258 labeled unanswerable. I compared a closed-book condition (Flan-T5-base with no retrieved context) against RAG with $K = 1, 3, 5$, and 10 retrieved passages.

The metrics were:

- **Exact match (EM)** and **token-level F1** against gold answers, after standard normalization.

- **Recall@K**: the proportion of questions whose gold passage appears among the top K retrieved passages.
- **Answerability detection** at $K = 5$: each output was classified as an answer or an abstention. Precision, recall, and F1 were computed for the answerable class.

All runs used a single Google Colab T4 GPU. The project reports one set of results per condition, with no variation across runs.

5. Results

5.1 Retrieval improved accuracy, from a very low base

Table 1 and Figure 1 summarize the results. Closed-book EM was 0.004, that is, 2 of 500 questions. With retrieval, EM rose to 0.034 at $K = 1$, 0.056 at $K = 3$, and 0.062 at $K = 5$ and $K = 10$. F1 rose from 0.027 to 0.091 at $K = 10$.

In relative terms, the gain at $K = 5$ is 15.5-fold. That figure should be read with care, because the baseline is two correct answers. The absolute gain is 5.8 percentage points of EM, and the 95% intervals for closed-book and $K = 5$ EM do not overlap (Table 1).

Table 1. Answer quality and retrieval by condition ($n = 500$).

Condition	EM	95% CI for EM	F1	Recall@K
Closed-book	0.004	0.001–0.014	0.027	n/a
RAG, $K = 1$	0.034	0.021–0.054	0.058	0.116
RAG, $K = 3$	0.056	0.039–0.080	0.081	0.194
RAG, $K = 5$	0.062	0.044–0.087	0.089	0.215
RAG, $K = 10$	0.062	0.044–0.087	0.091	0.310

Note. Confidence intervals are Wilson score intervals, which I computed from the reported proportions ($n = 500$).

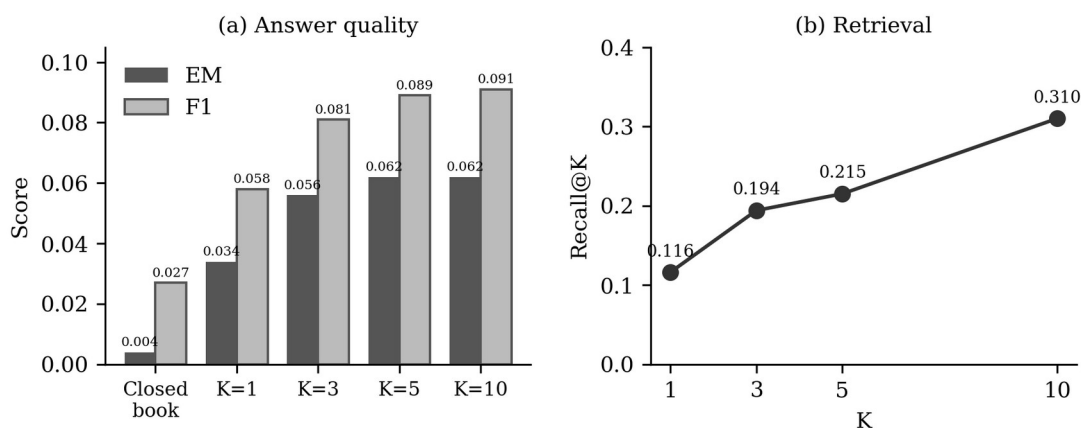


Figure 1. (a) Exact match and F1 for the closed-book baseline and RAG with K retrieved passages. (b) Recall@ K of the DPR retriever.

5.2 Retrieval usually missed the source passage

Recall@1 was 0.116 and Recall@10 was 0.310. Even with ten passages, the source paragraph was absent from the context for most questions.

These values are not directly comparable to the retrieval accuracy that Karpukhin et al. (2020) report. Their measure counts a hit when any retrieved passage contains the answer string, and it is computed over a 21-million-passage Wikipedia collection. Their results do, however, show that SQuAD is difficult for DPR. Even a DPR trained on SQuAD reached 63.2% top-20 accuracy on SQuAD, below the 68.8% of the lexical baseline BM25. The authors suggest two reasons. SQuAD annotators wrote questions after seeing the passage, which creates high lexical overlap that favors BM25, and the data come from only 500+ Wikipedia articles (Karpukhin et al., 2020). My retriever was trained on Natural Questions and never saw SQuAD, so a low recall is not surprising.

5.3 The system never abstained

At $K = 5$, no output was an abstention (Table 2). All 258 unanswerable questions received an answer string. Precision for the answerable class was $242/500 = 0.484$, recall was 1.000, and F1 was 0.652.

The F1 of 0.652 looks moderate, but it comes entirely from answering everything. A system with no ability to detect unanswerable questions obtains the same score.

Table 2. Answerability at $K = 5$.

Gold label	Predicted: abstain	Predicted: answer
Answerable ($n = 242$)	0	242
Unanswerable ($n = 258$)	0	258

5.4 Wrong answers looked like right ones

Table 3 shows three examples from the evaluation slice. In the first, retrieval enabled a correct extractive answer that the closed-book model missed. In the second, both conditions gave a wrong answer. In the third, the question is labeled unanswerable, yet both conditions gave a specific number. Nothing in the form of the output distinguishes the first case from the third.

Table 3. Example outputs.

Question	Gold answer	Closed-book	RAG ($K = 5$)
Cascade type when complement proteins bind microbes?	catalytic cascade	cytochrome c cascade (wrong)	catalytic (correct)
Main defense mechanism of bacteria?	restriction modification system	prokaryote (wrong)	antimicrobial peptide (wrong)
How many beds at Maria Memorial?	Unanswerable	84	78

Note. Questions are shortened as they appeared in my project materials.

6. Discussion: What This Means for Learners

The results above are measurements of one system. This section is interpretation. It connects those measurements to prior research on reliance, and none of it was tested with learners in this study.

6.1 An answer to every question removes a cue

Suppose a tutor built like this system were placed in front of students. At $K = 5$, the source paragraph was among the retrieved passages about one time in five ($\text{Recall}@5 = 0.215$), and the answer was exactly correct for about one question in sixteen ($\text{EM} = 0.062$). Yet every question received an answer, and the system returns only the generated text, with no indication of confidence. For the 258 questions with no supported answer, every output was unsupported, and none was marked as such.

A learner using such a system has no behavioral cue for when to trust it. Prior work suggests learners are unlikely to supply that judgment themselves. Users tend to overestimate the accuracy of LLM answers presented with default explanations (Steyvers et al., 2025). They rely on outputs whether or not certainty is marked (Zhou et al., 2024). They overrely when a model is convincingly wrong (Si et al., 2024). In one qualitative study, students who were struggling had more difficulty than their faster peers in setting aside incorrect AI suggestions, and often overestimated their own performance (Prather et al., 2024). The answer to “Maria Memorial has how many beds?” (78) is exactly the kind of confident, checkable-looking specific that a learner might copy into an assignment.

Retrieval may make this worse, not better. A RAG tutor can truthfully say that it grounds answers in course materials. In this system, the source paragraph was missing from the context for most questions, even at $K = 10$ (Recall@10 = 0.310). The retrieval step was present in every case, but the grounding it implies was not.

6.2 Aggregate metrics hid the failure

If I had reported only EM and F1, the headline would have been a 15.5-fold improvement from retrieval. The abstention failure appears only when answerable and unanswerable questions are scored separately. Even then, the detection F1 of 0.652 looks like partial success until one sees the confusion matrix.

This mirrors a point from the trust literature. How well people can discriminate correct from incorrect AI output is a different quantity from their average confidence (Steyvers et al., 2025). The same is true for systems. For an educational deployment, “how often is the tutor right?” is not enough. An evaluation should also ask how often the tutor answers when it should not, and should report that number separately.

6.3 Abstention as a teaching move

For a tutor, “I don’t know” need not be a dead end. A response such as “I could not find this in your course materials” tells the learner something true about the system’s limits and about where to look next. Research on uncertainty expression suggests the form of such messages matters: first-person uncertainty reduced agreement with the system and increased accuracy, while uncertainty expressed from a general perspective had weaker, non-significant effects (Kim et al., 2024).

Whether this transfers to learners is an open question. Two concerns deserve testing. First, abstention could lower learners’ trust in correct answers along with incorrect ones. Second, confidence information alone did not improve decisions in earlier work (Zhang et al., 2020). There is also a second reason abstention may matter in education. Bastani et al. (2025) found that safeguards, especially teacher-designed hints in place of answers, largely mitigated the harm to learning seen with unrestricted access. A tutor that declines to answer when it has no basis may share some of that benefit, but this is a hypothesis, not a finding.

7. A Proposed Study

To move from argument to evidence, I propose a between-subjects experiment with students. Participants would answer reading-comprehension questions with help from one of two versions of the tutor: the current system, or a version that abstains when retrieval confidence is low.

The abstaining version could be built with a calibrator in the style of Kamath et al. (2020) or with an explicit refusal instruction in the prompt. Its abstention accuracy would first be measured on held-out SQuAD 2.0 questions.

The question set would include answerable questions, unanswerable questions, and answerable questions where the tutor is wrong. The main outcome would be appropriate reliance: agreement with the tutor when it is correct and disagreement when it is not. Secondary outcomes would be discrimination between

correct and incorrect tutor outputs (Steyvers et al., 2025), self-reported trust, and performance on a later test without the tutor, following the design logic of Bastani et al. (2025). Given evidence that AI literacy moderates how trust translates into reliance (Pitts et al., 2026), I would measure AI literacy and test it as a moderator.

8. Limitations

This was a course project run on a single GPU. The results come from one seeded 500-question sample, and no variation across runs or samples is reported.

- **Small counts.** The closed-book baseline rests on two correct answers, so relative gains are unstable. Differences among the retrieval conditions (for example, EM 0.056 at $K = 3$ versus 0.062 at $K = 5$) fall within overlapping intervals.
- **Small, off-the-shelf components.** The retriever was used off the shelf and never trained on SQuAD, and the generator is small. Stronger or fine-tuned systems would likely score higher, and some might abstain. The finding is not that RAG systems cannot abstain. It is that a standard pipeline assembled without an explicit abstention mechanism did not, and that the headline metrics did not reveal this.
- **Abstention mechanism.** Abstention was not explicitly supported by the prompt or the decoding procedure. I cannot separate the contributions of the prompt, the decoding settings, and the model to the failure.
- **Evaluation data.** The collection was built from SQuAD contexts, not from course materials, and SQuAD questions are not student questions.
- **No learners.** No learners were involved. Every claim in Section 6 about learners is an inference from prior studies.

9. Conclusion

A retrieval-augmented question-answering system assembled from standard components improved over a no-retrieval baseline. It also answered all 258 questions it had no basis to answer, in the same form as its correct answers. For education, the second result matters more than the first. A tutor that never says “I don’t know” gives learners nothing to calibrate against, and the metrics most often reported would not show the problem. I suggest that educational AI be evaluated on whether it abstains when it should, and that abstention be studied as part of tutor design rather than as a technical afterthought.

10. Declarations

Origin of this work. The system and all results in Sections 3 to 5 come from my final project for CSCI E-222, Foundations of Large Language Models, at Harvard University (Spring 2026). This paper reframes that project around an education question. The literature review, the confidence intervals, and Sections 6 and 7 are new.

Use of AI tools. I used Claude (Anthropic), an AI assistant, in preparing this manuscript. It checked the reported numbers against my project materials, verified references against publisher records, computed the confidence intervals, drafted text, and produced Figure 1 from my reported results. I defined the research question, built and ran the system, chose the framing, reviewed and revised the text, and take full responsibility for the content.

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